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General Comment

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Attachments

AI Request for National Science Foundation

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Response to the National Science Foundation Request for Information on the Development of an Artificial Intelligence (AI) Action Plan -- Beacon College, March 12, 2025

The landscape of higher education is undergoing significant shifts, driven by changing demographics, escalating costs, and increasing scrutiny over the value of an undergraduate degree. The Beacon College community recognizes the unique position we hold in this evolving environment. Unlike many of our competitors, Beacon is dedicated to serving a distinct profile of collegiate learners — students who exhibit learning and attention issues (heretofore labeled “neurodiverse learners”). This student population requires a tailored approach that prioritizes developmental themes and structures designed specifically to meet their needs. As a result of our efforts, Beacon College has experienced notable outcomes in persistence, degree completion, and post-graduate placements. However, the current estimates are that one in seven students is neurodiverse, underscoring the growing challenges faced by this population and the need for ongoing innovation in support services.

This document outlines the Beacon College Triadic Developmental Model, our current use of AI to support neurodiverse students, and our strategic vision for AI’s role in the future. It is our hope that the development of AI technologies will reflect the needs of neurodiverse learners, a talent pool that remains underutilized in the workforce.

The Triadic Developmental Model at Beacon College

The Triadic Developmental Model aims to foster successful student engagement, which, in turn, promotes persistence (students' ability to continue working towards their academic goals) and retention (the college's ability to support student persistence). Student engagement is defined as active participation in academic, campus, and social activities that students perceive as supportive of their learning and development, leading to positive outcomes for both the students and the institution.

Research shows that complex interventions are often more effective than isolated skill training. Accordingly, the Triadic Model combines college preparation, academic engagement, and holistic development, supported by integrated wrap-around services. These services include Transition Counselors, Learning Specialists, and Student Life Professionals who provide ongoing support throughout the college journey. Unlike many schools with separate transition, academic, or social support programs, Beacon synthesizes these services into a comprehensive, tailored support system.

Our approach is grounded in evidence-based practices, emphasizing executive function and social-emotional skills development, alongside academic support. By focusing on individualized coaching, skill-building, and experiential learning, we aim to help students cultivate the independence required to thrive in both college and their careers. Through fostering self-advocacy, resilience, executive function, and life skills, our model sets the foundation for lifelong success.

Current and Strategic Use of AI for Neurodiverse Students

The implementation of the Triadic Model at Beacon College has significantly improved our students' experience and their success. By combining the roles of college preparation, academic engagement, and holistic development, the Triadic Model has made interventions more explicit, individualized, comprehensive, and differentiated, while also allowing for adjustments in intensity based on student needs. Here are the three elements of the model, briefly described.

1. College Preparation and Transition–

As part of our mission to provide a supportive and accessible higher education experience, the Center for College Readiness at Beacon College helps bridge the gap between high school and higher education for students with learning differences. Traditional transition pathways often fail to address the unique needs of this population, resulting in lower college retention and graduation rates. Beacon's comprehensive preparatory programs are designed to equip students with the skills, confidence, and strategies necessary to achieve their academic goals while readying them for their professional life.

Beacon offers a variety of programs for prospective college students through the Center for College Readiness. These include multi-week residential programs for high school students, virtual coaching, dual enrollment, introduction to college academics, first year college curriculum that is online with added support, and family education. We also offer an online coaching certificate for coaching neurodiverse adolescents through this Center. The Center also provides consulting services and resource-sharing for high schools and colleges looking to improve their ability to support neurodiverse students. We are proud to offer the certificate and coaching services, as they provide us the opportunity to disseminate strategies for supporting neurodiverse students to a larger number of professional stakeholders.

2. Academic Engagement –

Within the academic domain, it is important to align academic achievement (e.g., grades, rate of progress) with learning and developmental domains and dimensions (i.e., behavioral, emotional, cognitive). To achieve this goal, The Center for Student Success assigns each Beacon College student a Learning Specialist. These educational professionals provide one-on-one coaching, mentoring, and support for students in their academic journey. Their approach is centered on academic engagement in the behavioral, emotional, and cognitive realms. Learning Specialists also provide support for families throughout their loved ones' college experience. The goal of their approach is to facilitate students' success, thus increasing their academic confidence.

3. Holistic Development –

A large part of a student's college experience, and an important part of their development occurs outside of the classroom. Beacon College students must develop holistic skills they will need to succeed in college and to live an abundant life. To achieve these outcomes, our Student Affairs team offers clubs, college athletics, fitness and recreational

activities, and a multitude of other programs for students that enrich their college experience. Further, participation in these activities helps students develop holistic skills such as leadership, interpersonal effectiveness, and social responsibility.

Current and Strategic use of AI for Neurodiverse Students

Beacon College is committed to harnessing AI to enhance access to resources and personalized learning strategies for neurodiverse students. AI-driven tools have the potential to support executive function coaching, academic accommodations, and overall accessibility. However, these tools must be designed inclusively to ensure they meet the needs of this student population.

Here are examples of how AI is currently being used at Beacon College, along with strategic ideas for future development:

1. College Preparation and Transition –

Beacon College currently uses data analytics to predict student retention and success. While significant strides have been made in understanding predictors for neurotypical students, the same is not yet true for neurodiverse learners. AI can help develop predictive models that identify factors influencing success for neurodiverse students. By using AI, we can also uncover potential challenges early and provide tailored interventions before these challenges become barriers. AI could be used to assess students' strengths and challenges, helping us create personalized strategies to facilitate smoother transitions to college.

2. Academic Engagement –

Beacon College employs AI tools such as NotebookLM and ChatGPT to support students' academic engagement. These tools help students, faculty, and Learning Specialists rephrase assignments, break tasks into manageable segments, and create multi-modal versions of content to fit students' individual learning styles. Additionally, technologies like GoblinTools assist with time management and task organization, which are essential skills for neurodiverse students. Tools such as ReadAloud and Kurzweil use AI to provide text-to-speech and speech-to-text capabilities, enabling students to access content in alternative formats that suit their needs.

Future advancements in AI will enable Beacon faculty to better implement multi-modal teaching strategies, analyze student performance, and automate administrative tasks. This will free up more time for one-on-one interactions with students, providing more personalized support and enhancing overall student learning.

3. Holistic Success –

Beacon College uses AI-powered technologies to support students in preparing for job interviews. These tools provide feedback on interviews, helping students refine their skills. However, AI technology tailored to neurodiverse students, offering real-time feedback on emotional regulation, organization, and interpersonal skills, would be highly beneficial. Generative AI could help students build confidence and develop strategies to succeed in the competitive job market.

Additionally, the role of families in supporting neurodiverse students is crucial, yet many families struggle to understand how best to provide that support. AI could help by enabling simulations where parents practice conversations with their children or with professionals, fostering a deeper understanding of how to best support their loved ones during their educational journey.

4. Professional Development for faculty and staff –

Faculty and staff at Beacon College need ongoing training in several key areas, including multi-modal teaching, learning and assessment strategies, and differentiated instruction for neurodiverse students. AI tools can enhance professional development by simulating real-world scenarios through case studies, allowing faculty to engage in role-playing and problem-solving exercises. Adaptive AI technologies will also offer personalized instruction, real-time feedback, and guidance based on individual needs, improving the quality of teaching and support across the campus.

Looking Ahead: AI in Support of Neurodiverse Learners

At Beacon College, we firmly believe that neurodiverse students represent an underutilized resource in the American workforce. They bring unique skills and perspectives that are invaluable in enriching the workforce, and their success is vital to a strong and dynamic economy. As we continue to explore the potential of AI, we are excited about its ability to enhance the support we offer to neurodiverse students, as well as to professionals in higher education and the workforce.

We welcome further dialogue with policymakers, educators, and technology leaders to ensure AI is implemented in ways that empower students with learning differences to achieve their full potential.